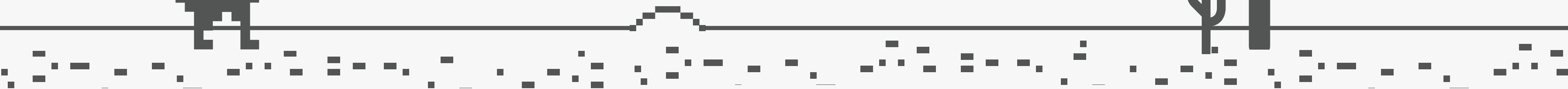




ANALYZING GAMING ENGAGEMENT PATTERNS: A STATISTICAL FRAMEWORK FOR EVALUATING GENRE PREFERENCES AND PLAYER BEHAVIOR



P R E S E N T E R : K I N G D H E N I E L N . C O N C E P C I O N





1 INTRODUCTION



This study establishes a systematic statistical framework using Playnite to transform raw historical playtime data from a personal digital library into verifiable behavioral insights. By applying non-parametric tests like Spearman's Rank Correlation, the research seeks to identify core engagement drivers and power-law patterns to provide a reliable roadmap for predicting future media consumption.





PROBLEM STATEMENT



Digital gaming libraries accumulate vast amounts of personal data, yet these records often remain unexamined at a personal level. Without a structured analytical approach, it is difficult to identify core engagement drivers or understand how session frequency impacts total time investment in a highly skewed data environment. This study addresses this gap by applying a quantitative statistical framework to model personal gaming behavior and demonstrate that digital consumption follows predictable, evidence-based patterns.



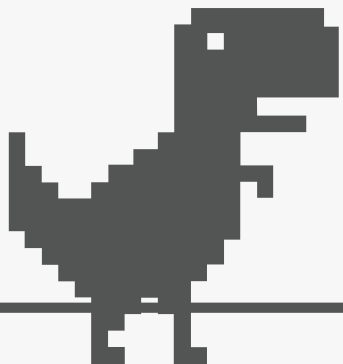


RESEARCH QUESTIONS

General Research Question: How can a quantitative statistical framework be utilized to analyze and model the engagement patterns and behavioral trends within a personal digital gaming library?

Specific Research Questions:

1. What is the distribution profile of the gaming library regarding Time Played Hours and Play Count, and how does it show the difference between high-engagement titles from the rest of the collection?
2. Is there a statistically significant relationship between session frequency & total duration and what is the strength of this association?
3. How do gaming preferences evolve across different platforms and categories, and which genres currently lead engagement compared to historical data?





OBJECTIVES OF THE STUDY

General Objective: To develop and apply a quantitative statistical framework for analyzing and modeling player engagement patterns and behavioral trends within a personal digital gaming library.

Specific Objectives:

- 1.To characterize the distribution profile of the gaming library by calculating descriptive statistics for Time Played Hours and Play Count, identifying the presence of power-law patterns and high-engagement "outlier" titles.
- 2.To evaluate the statistical relationship between session frequency and total play duration by applying Spearman's Rank Correlation to determine the significance and strength of the association.
- 3.To analyze the temporal evolution of gaming preferences by comparing historical genre and platform data against recent engagement metrics to identify shifts in player interest.





2 REVIEW OF RELATED LITERATURE (RRL)



The literature review identifies a critical research gap in the analysis of personal, small-scale digital libraries, noting that current studies often prioritize "Big Data" over individual player habits. By examining the evolution of universal managers like Playnite, the review establishes the necessity of using formal statistical tests and data cleaning to ensure that playtime metrics accurately reflect high-quality engagement.

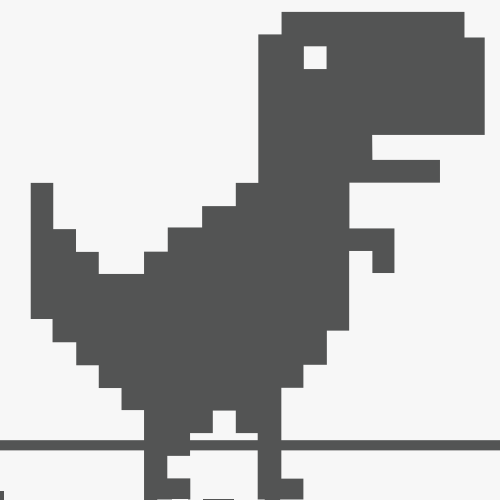




2 REVIEW OF RELATED LITERATURE (RRL)



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3 METHODOLOGY

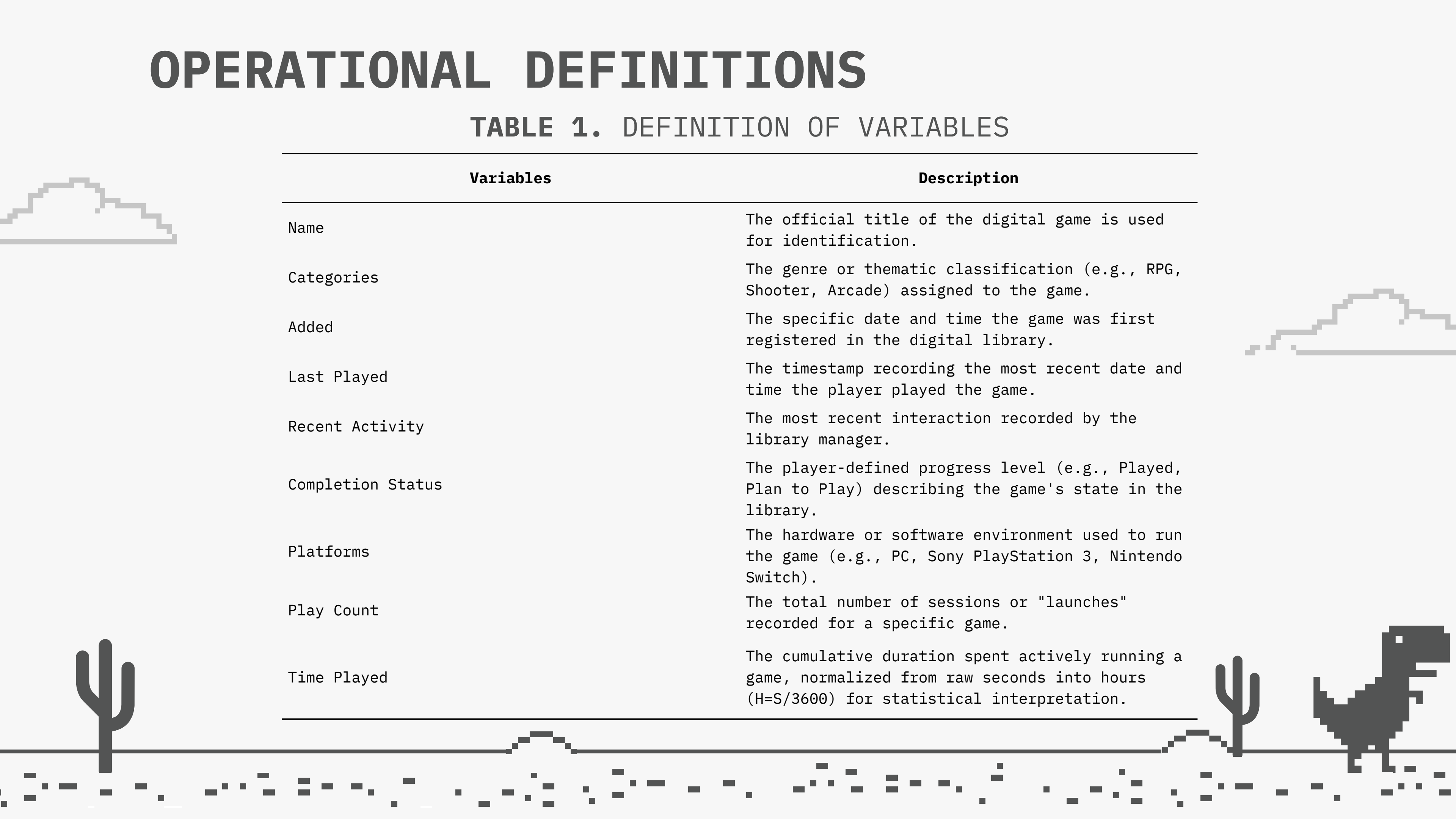
- This study follows a quantitative descriptive-correlational design using Exploratory Data Analysis (EDA) to examine localized gaming habits.
- The systematic process involves automated data collection, rigorous preprocessing, and the application of both parametric and non-parametric statistical tests.

PARTICIPANTS

- **Academic Background:** The subject is a 21-year-old fourth-year student currently pursuing a Bachelor of Science in Computer Science at National University - Manila.
- **Technical Proficiency:** Due to an academic specialization in computer science, the subject possesses an inherent technical proficiency in managing complex digital ecosystems.
- **Player Profile:** Beyond academic requirements, the subject is identified as a dedicated gaming practitioner and a collector of action figures.
- **Research Context:** This background provides the necessary context for the maintenance of a diverse and extensive digital library, serving as the primary data source for this investigation.

DATA COLLECTION METHODS

- **Aggregation Tool:** Data was collected using Playnite, an open-source video game library manager that aggregates titles from various services including Steam, Epic Games, and PlayStation.
- **Study Timeline:** The dataset covers several years of historical activity and was finalized on February 13, 2026.
- **Variables Collected:** The study utilized nine core attributes: Name, Categories, Added, Last Played, Recent Activity, Completion Status, Platforms, Play Count, and Time Played.
- **Frequency of Logging:** Data logging was performed automatically by the software, recording launch frequency and session duration without manual intervention.
- **Sample Size:** The final dataset consists of 79 unique game titles across various digital ecosystems.



OPERATIONAL DEFINITIONS

TABLE 1. DEFINITION OF VARIABLES

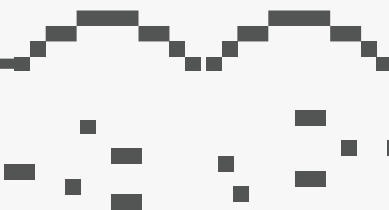
Variables	Description
Name	The official title of the digital game is used for identification.
Categories	The genre or thematic classification (e.g., RPG, Shooter, Arcade) assigned to the game.
Added	The specific date and time the game was first registered in the digital library.
Last Played	The timestamp recording the most recent date and time the player played the game.
Recent Activity	The most recent interaction recorded by the library manager.
Completion Status	The player-defined progress level (e.g., Played, Plan to Play) describing the game's state in the library.
Platforms	The hardware or software environment used to run the game (e.g., PC, Sony PlayStation 3, Nintendo Switch).
Play Count	The total number of sessions or "launches" recorded for a specific game.
Time Played	The cumulative duration spent actively running a game, normalized from raw seconds into hours ($H=S/3600$) for statistical interpretation.

EDA AND STATISTICAL TECHNIQUES

The study employs a quantitative descriptive-correlational design using **Exploratory Data Analysis (EDA)** to model gaming patterns:

- **Descriptive Statistics:** Summarizes central tendency and dispersion using Mean, Median, and Standard Deviation, alongside Skewness and Kurtosis to justify the use of non-parametric tests.
- **Spearman's Rank Correlation:** Utilized as the primary robust, non-parametric measure to evaluate monotonic relationships and address highly skewed playtime and session frequency data.
- **Pearson Correlation Coefficient:** Applied as a secondary linear metric to assess direct trends and verify consistent associations between session frequency and total duration.
- **Data Visualization Framework:** Leverages Matplotlib and Seaborn to generate histograms for frequency distribution, box plots for category variance, and scatter plots for regression trends.
- **Standardized Mapping & Normalization:** Implements seven core genre labels for consistent grouping and converts raw seconds into hours for professional reporting.

4 RESULTS



OVERALL DATASET OVERVIEW

The gaming library dataset consists of 79 total games, with the primary analysis focusing on a subset of 72 titles that hold a "Played" status and a play count greater than zero. These active games span six distinct platforms, including the Nintendo Switch, Nintendo Switch 2, PC (Windows), and Sony PlayStation 3, 4, and 5, while being organized into seven core categories: Action, Arcade, Fighting, RPG, Shooter, Simulation, and Survival. By filtering titles with confirmed session activity, this 72-game dataset serves as the standardized basis for all descriptive and correlational statistical measures presented in this report.

DESCRIPTIVE STATISTICS

TABLE 2. DESCRIPTIVE STATISTICS SUMMARY

Statistic	Time Played (Hours)
Mean	29.2
Median	3.81
Standard Deviation	82.96
Minimum	0
Maximum	570.58
25th Percentile	1.31
75th Percentile	13.8
Skewness	4.9
Kurtosis	27.44

DESCRIPTIVE STATISTICS

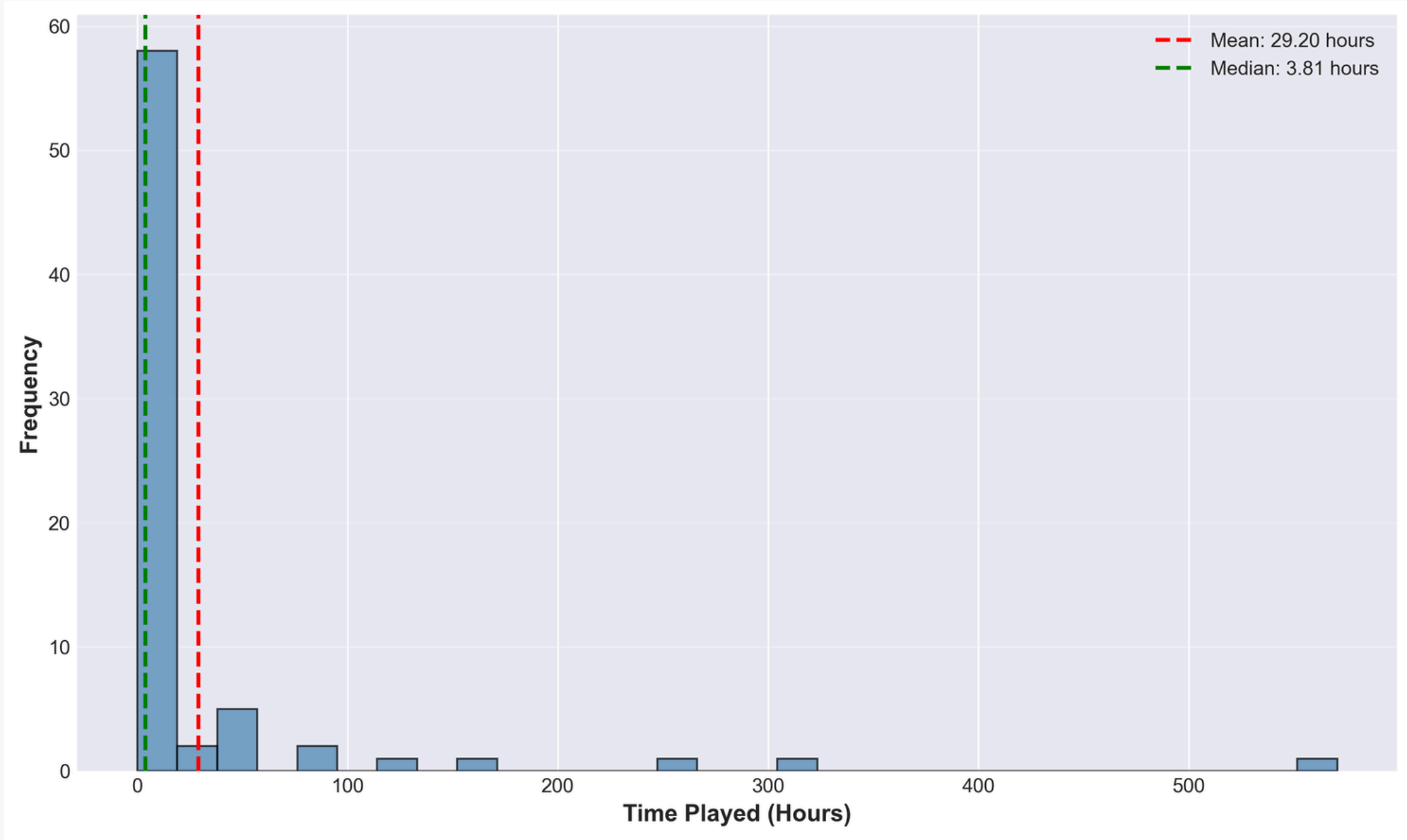


FIG. 1. HISTOGRAM OF TIME PLAYED HOURS

DESCRIPTIVE STATISTICS

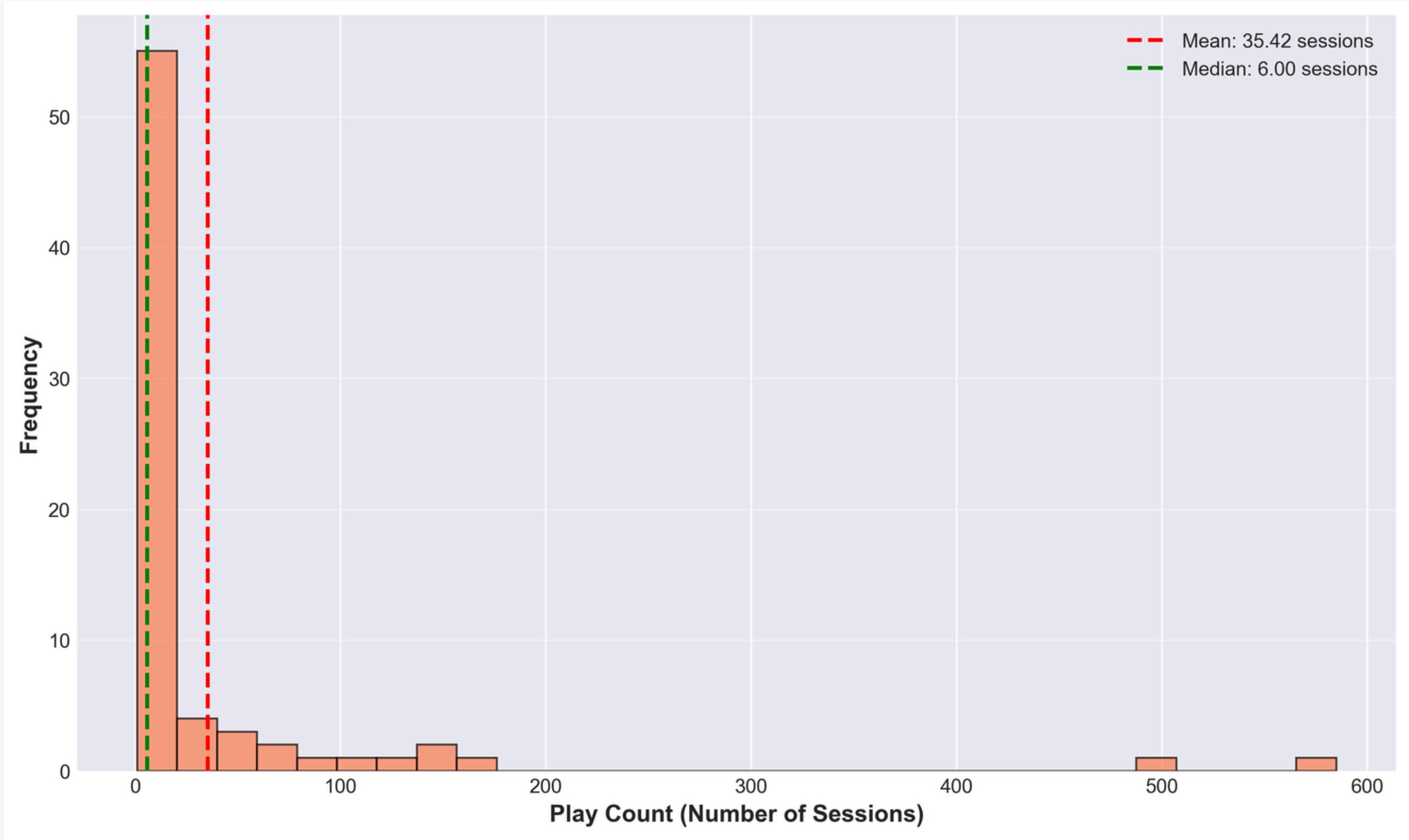


FIG. 2. HISTOGRAM OF PLAY COUNT

TIME-SERIES TRENDS

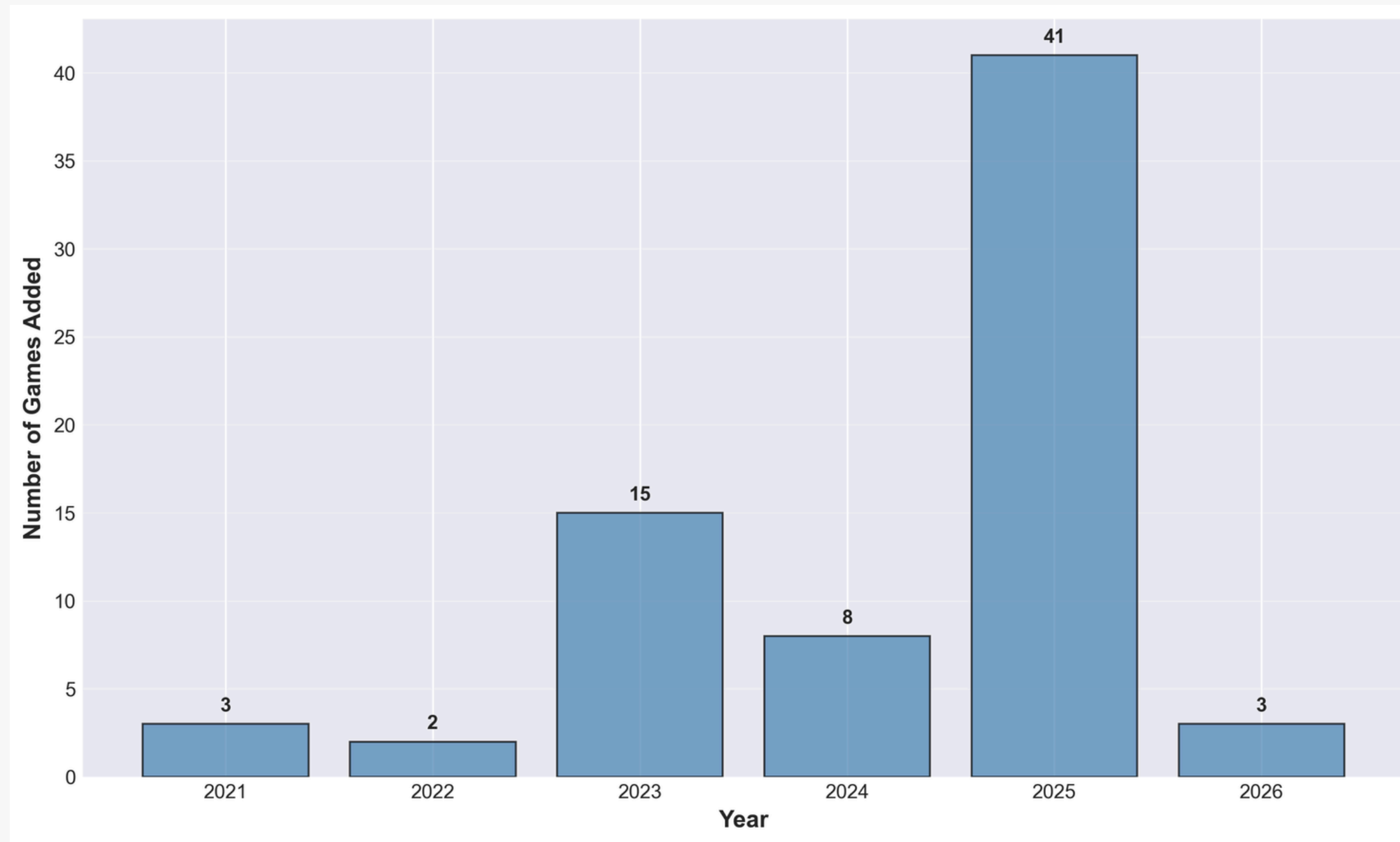


FIG. 3. GAMES ADDED PER YEAR (2021-2026)

TIME-SERIES TRENDS

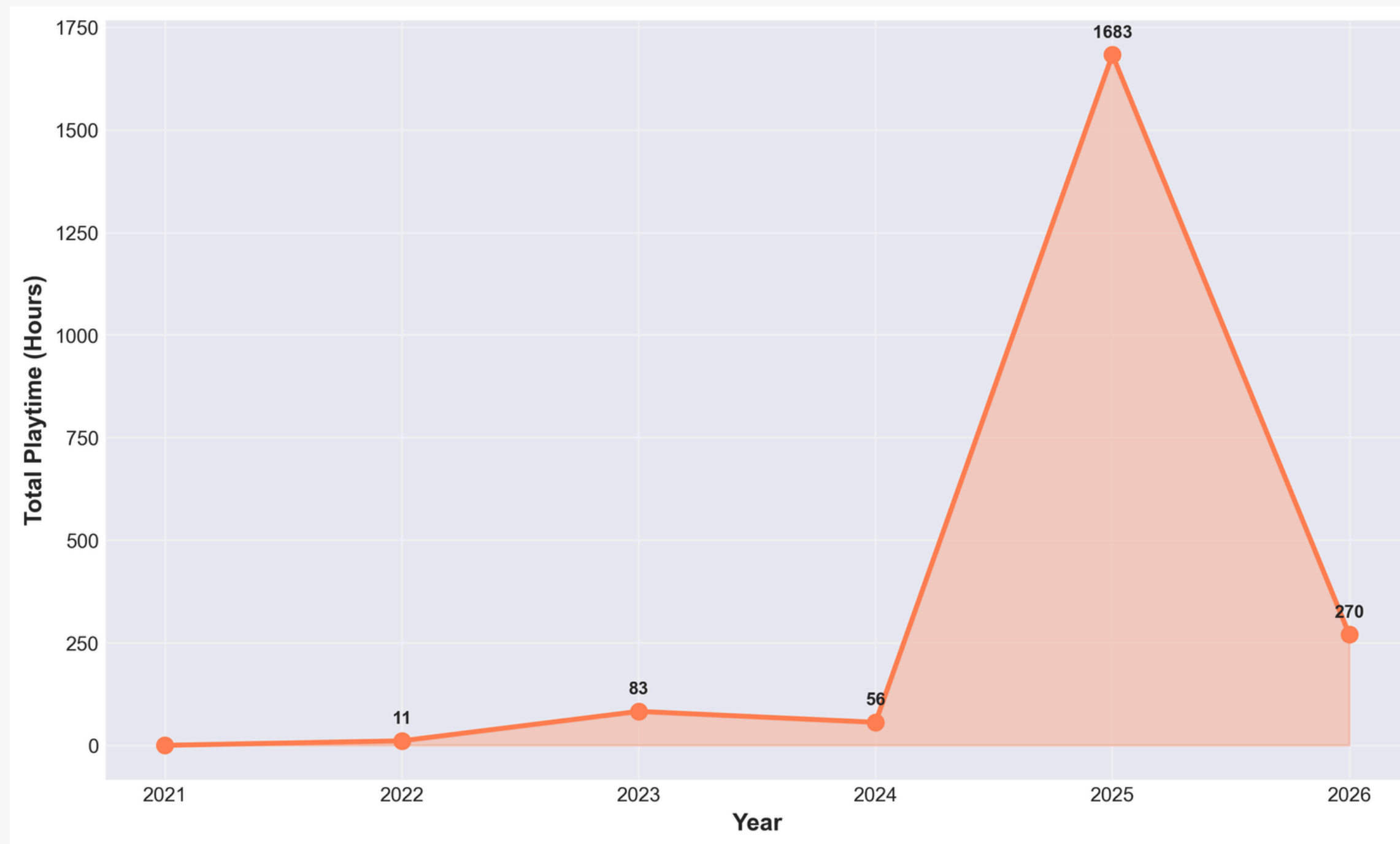


FIG. 4. TOTAL PLAYTIME PER YEAR (2021-2026)

CORRELATION ANALYSIS

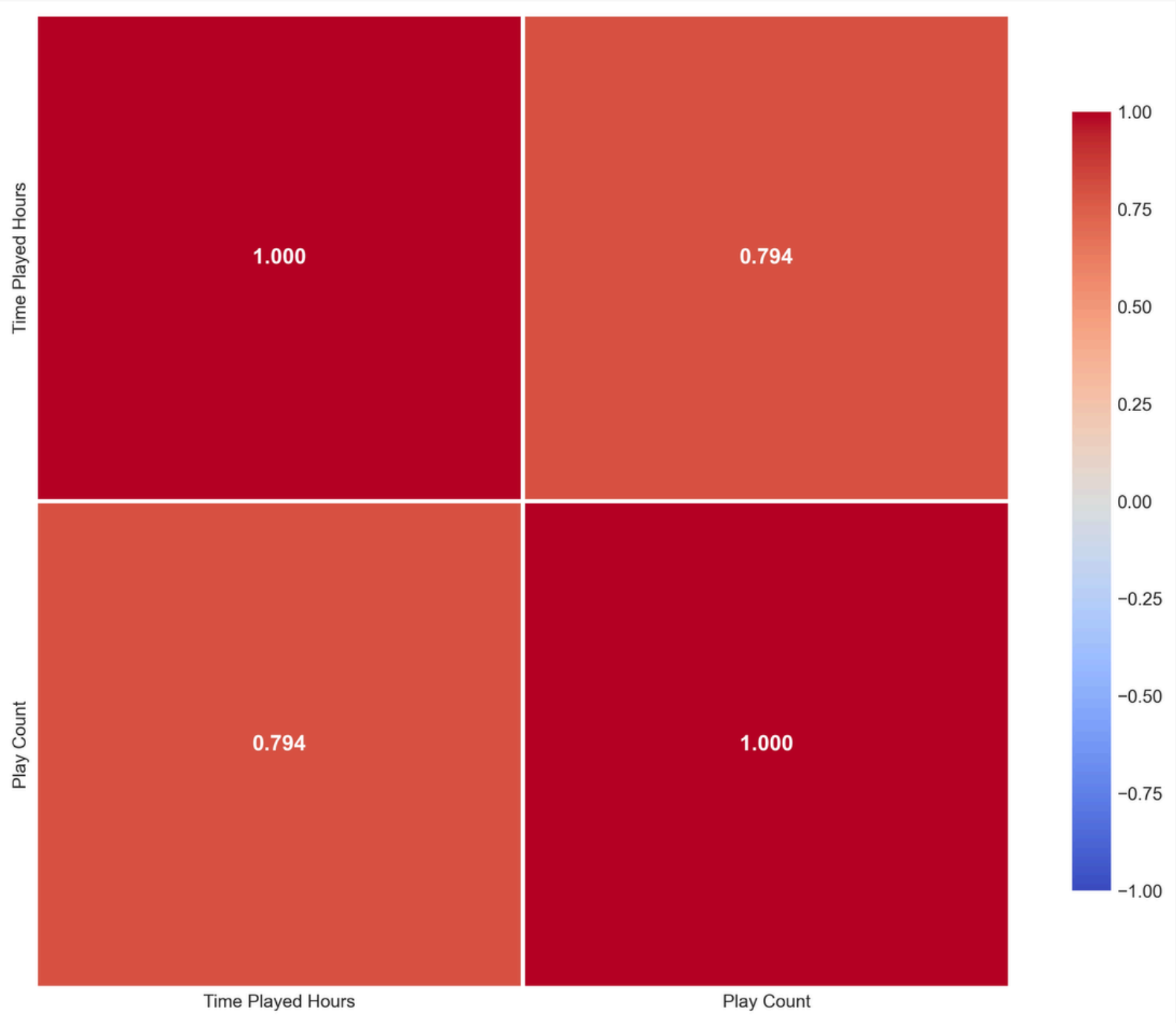


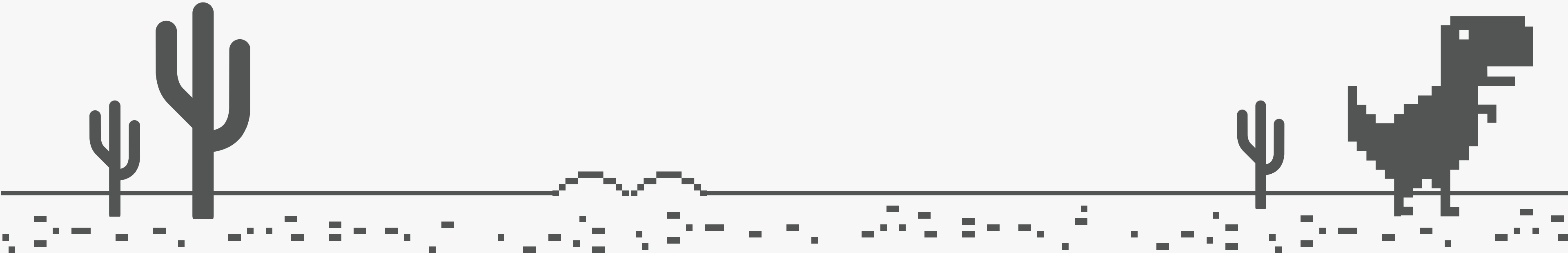
FIG. 5. CORRELATION MATRIX HEATMAP

CORRELATION ANALYSIS



TABLE 3. SUMMARY COMPARATIVE ANALYSIS

Variable Pair	Pearson	Spearman	p-value
Time Played Hours vs Play Count	0.7942	0.7942	< 0.001



CORRELATION ANALYSIS

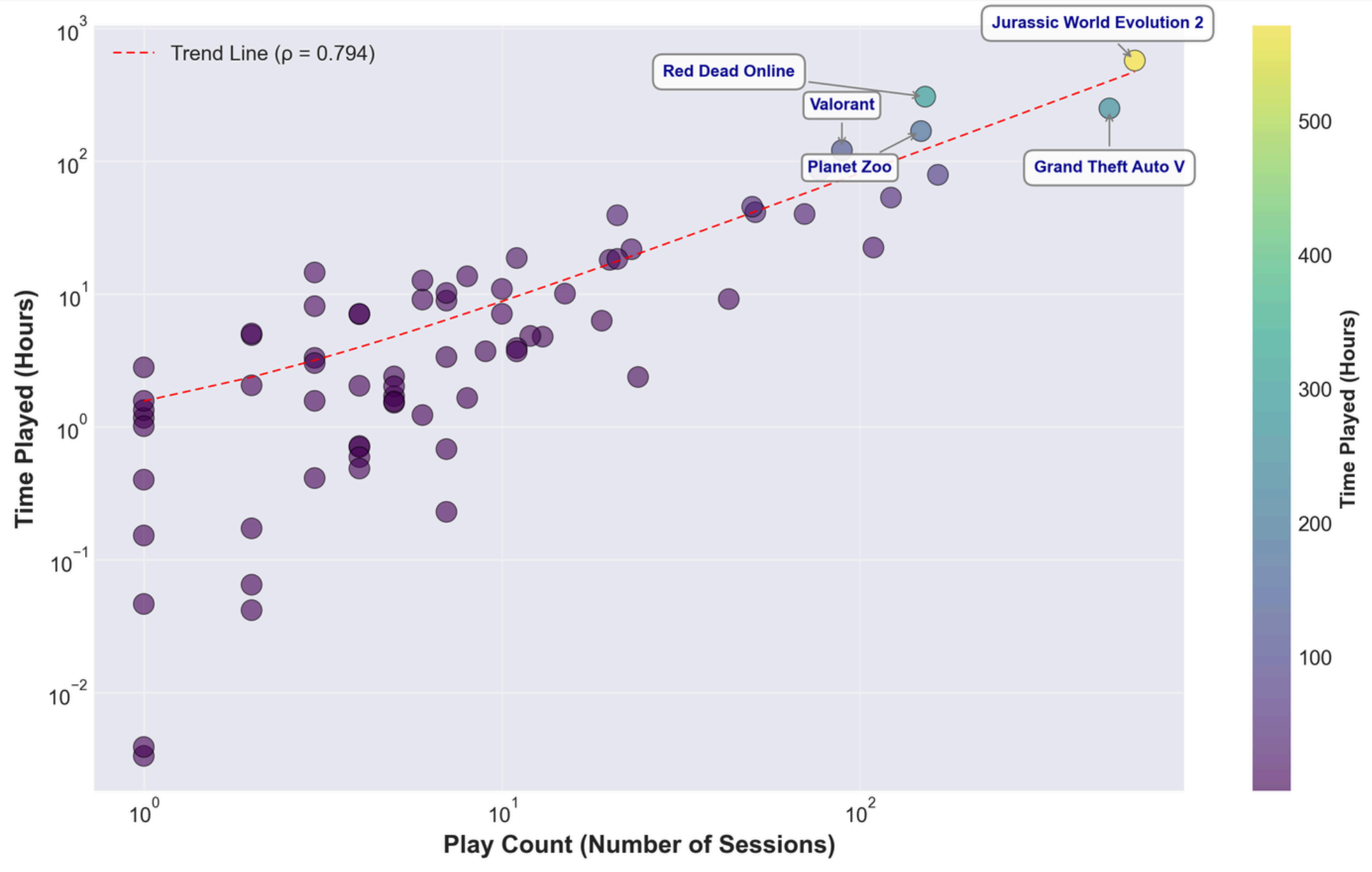


FIG. 6. SCATTER PLOT OF PLAY COUNT VS. TIME PLAYED HOURS

PLATFORM DISTRIBUTION ANALYSIS

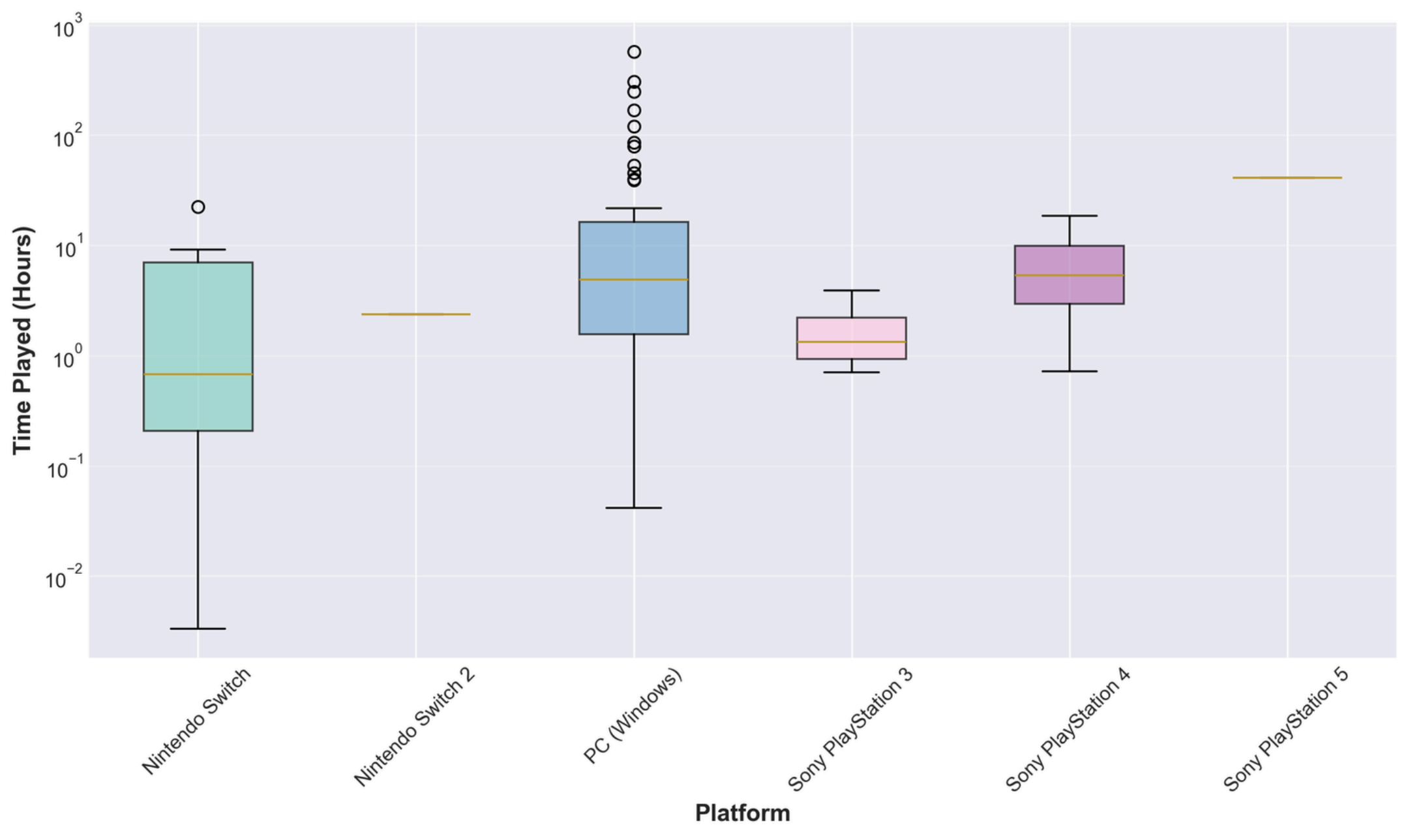


FIG. 7. PLATFORM DISTRIBUTION

CATEGORY ANALYSIS

TABLE 4. CATEGORY ANALYSIS SUMMARY

Categories	Number of Games	Total Hours Played	Avg Hours per Game	Median Hours	Total Play Sessions	Avg Play Sessions
Simulation	9	824.25	91.58	1.57	912	101.33
Action	16	763.96	47.75	8.22	874	54.62
Shooter	14	231.23	16.52	6.06	204	14.57
Survival	10	159.41	15.94	13.1	151	15.1
Arcade	12	79.5	6.63	1.6	230	19.17
RPG	4	24.7	6.17	1.19	135	33.75
Fighting	7	19.56	2.79	2.03	44	6.29

CATEGORY ANALYSIS

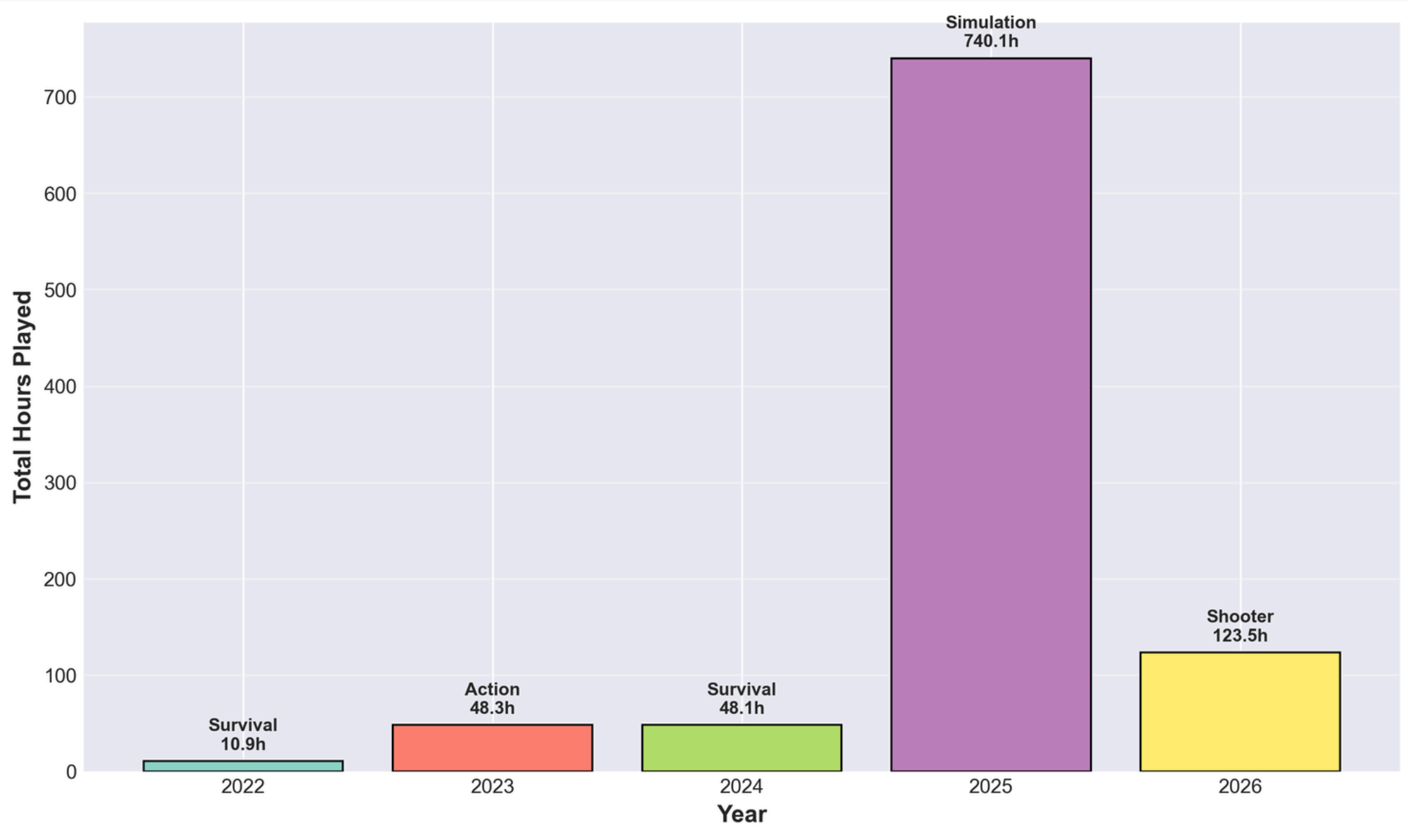


FIG. 8. TOP CATEGORY PER YEAR (2021-2026)

CATEGORY ANALYSIS

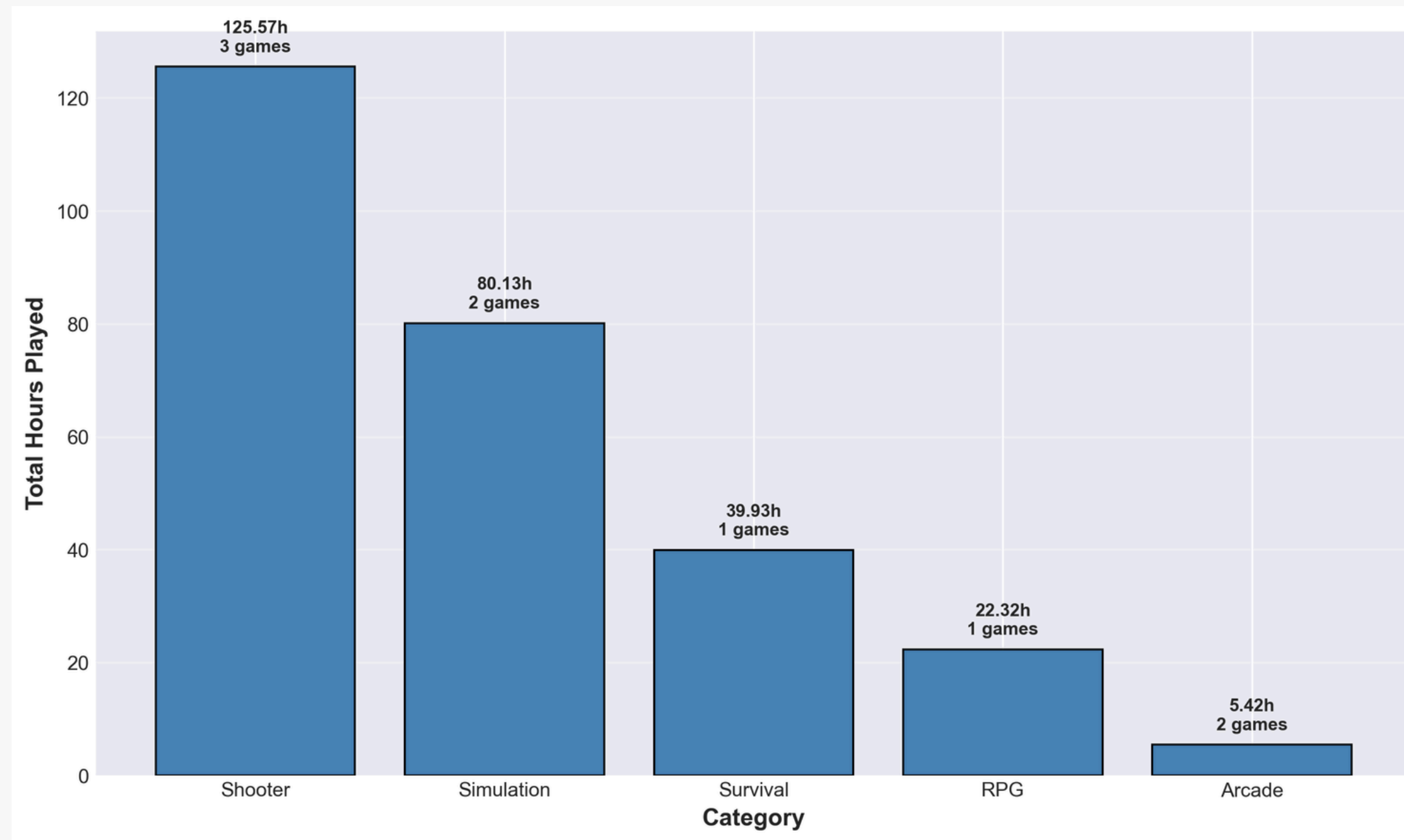


FIG. 9. PLAYTIME BY CATEGORY (PAST 30 DAYS)

SUMMARY OF STATISTICAL ANALYSIS

TABLE 5. SUMMARY OF STATISTICAL ANALYSIS

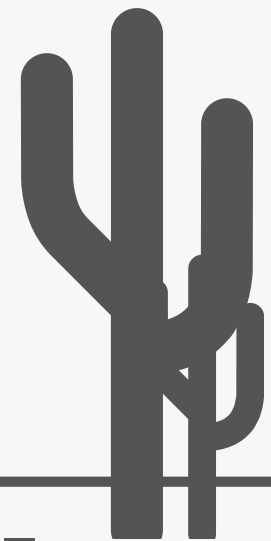
Component	Result
Sample size (played games)	72
Time Played (Hours) â€™ Mean	29.2
Time Played (Hours) â€™ Median	3.81
Standard Deviation (Time Played Hours)	82.96
Play Count â€™ Mean	35.42
Play Count â€™ Median	6
Standard Deviation (Play Count)	94.07
Spearman (Time Played vs Play Count)	0.7942
p-value	< 0.001
Significance (p = < 0.05)	Significant
Correlation interpretation	Strong positive correlation
Top category (overall by total hours)	Simulation (824.2 h)
Top category (past 30 days)	Shooter (125.6 h)



KEY VISUALIZATIONS SUMMARY

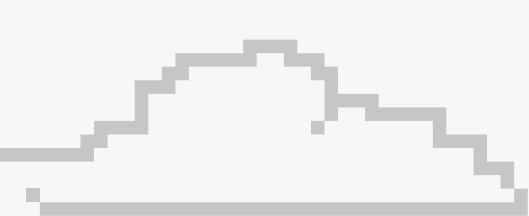
The research findings are consolidated through nine figures and four tables to illustrate player engagement patterns:

- **Distribution Profile:** Histograms reveal a right-skewed pattern where outliers reach up to 570.58 hours and 585 sessions.
 - **Temporal Peaks:** Both library growth (41 games) and player engagement (1,683 hours) reached their maximum in 2025.
 - **Correlation Strength:** A strong relationship between frequency and duration is confirmed by $p = 0.7942$ and $p < 0.001$.
 - **Genre Evolution:** Analysis shows a shift from Survival (2022) to Simulation (2025) and currently to Shooter (2026).
 - **Historical Leaders:** Simulation (824.2h) and Action (764.0h) remain the overall top categories in the dataset.
- 





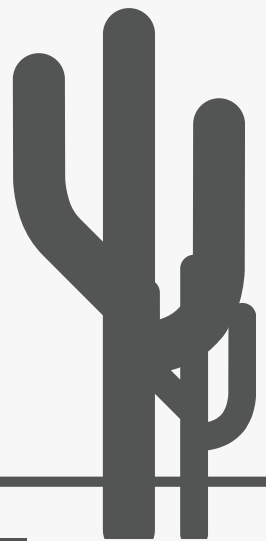
5 DISCUSSION

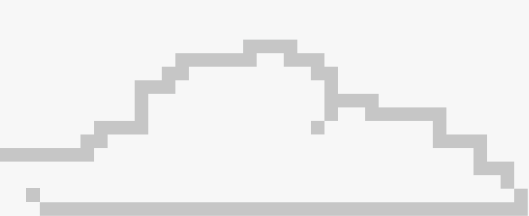


INTERPRETATION OF RESULTS

The analysis reveals distinct patterns in player engagement and consumption habits:

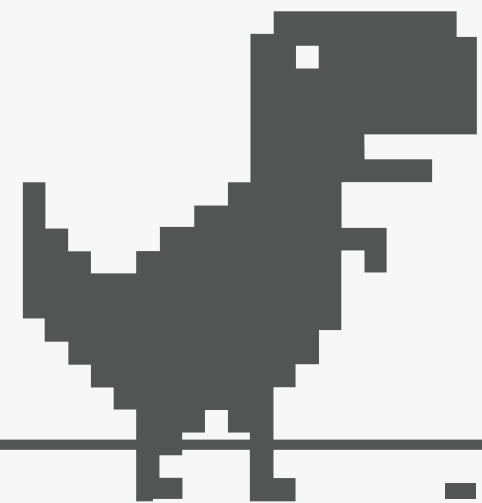
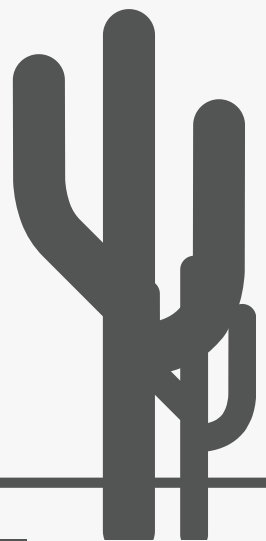
- **Selective Engagement:** A few titles like Jurassic World Evolution 2 account for most playtime, pulling the mean (29.20 hrs) far above the median (3.81 hrs).
- **Power-Law Distribution:** Engagement follows a right-skewed pattern where a small number of games dominate total library attention.
- **Engagement Loop:** A strong correlation ($p = 0.7942$) proves that frequent game launches directly lead to higher total time investment.
- **Temporal Peaks:** Activity is inconsistent, peaking significantly in 2025 (1,683 hours), likely due to academic breaks or major releases.
- **Genre & Platform Shifts:** While Simulation is the historical leader (824 hrs), there is a recent tactical shift toward Shooters.
- **Platform Preference:** The PC (Windows) platform hosts the highest number of high-playtime outliers compared to consoles.

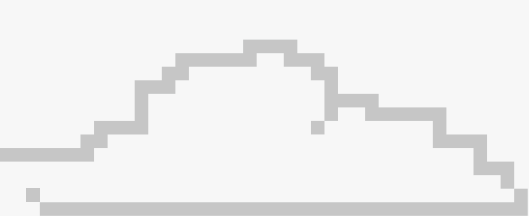




COMPARISON TO RELATED WORK

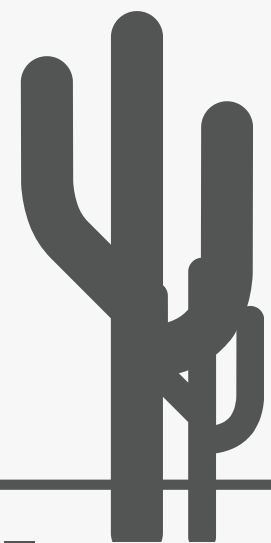
The findings from this study mirror broader digital consumption research by confirming that personal gaming behavior follows a predictable power-law distribution where a few items dominate total engagement. Furthermore, the strong correlation between play count and duration, alongside activity fluctuations driven by academic schedules, aligns with established patterns in student media habits and high-loyalty technology platforms.

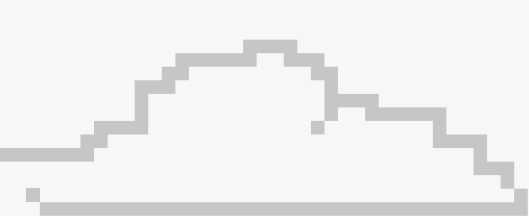




LIMITATIONS

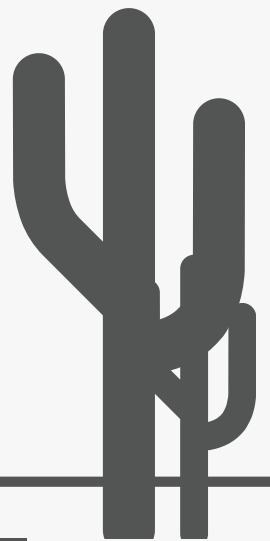
- **Small Sample Size:** Analysis is restricted to a single player's library of 72 games, representing individual habits rather than general trends.
- **Tracking & Bias:** Potential software errors or untracked activity via Playnite may affect data accuracy.
- **Missing Metadata:** The use of "Uncategorized" labels for missing genre data may influence category-based precision.
- **Temporal Scope:** The five-year snapshot (2021–2026) may not capture long-term longitudinal shifts in preferences.
- **Platform Bias:** Differences in how platforms integrate with the tracking software could bias results toward specific systems.
- **Quantitative Focus:** The study lacks qualitative context, such as social influences or personal game enjoyment

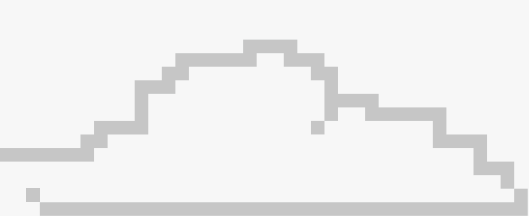




RECOMMENDATIONS AND FUTURE WORK

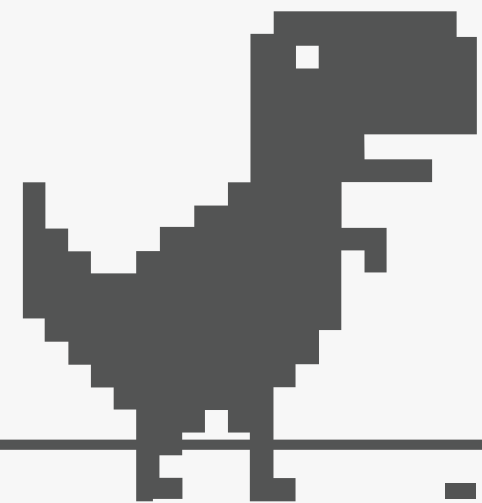
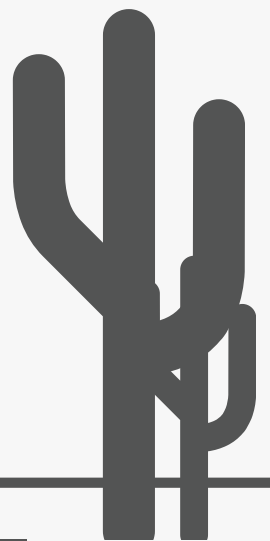
- **Suggestions for Students:** Future research should expand sample sizes, include additional variables like ratings and reviews, conduct longitudinal tracking, and perform systematic cross-platform hardware comparisons.
 - **Alternative Methodologies:** Future studies could implement machine learning for predictive modeling, utilize social network analysis for community trends, compare demographic groups, and integrate qualitative methods like surveys and interviews.
- 





5 CONCLUSION

This research successfully established a robust analytical framework using non-parametric tools to prove that player engagement follows a predictable power-law pattern driven by specific session-frequency feedback loops. The findings demonstrate that while historical dominance belongs to Simulation and Action genres, engagement is highly concentrated in specific outlier titles and fluctuates according to external factors like academic schedules.



THANK YOU

